branes from buckling. The only thing he can think of to do the job is bulk fields. Whenever a confining brane starts to bend, bulk fields must somehow exert a force on it, pushing it back into its proper, straight shape.

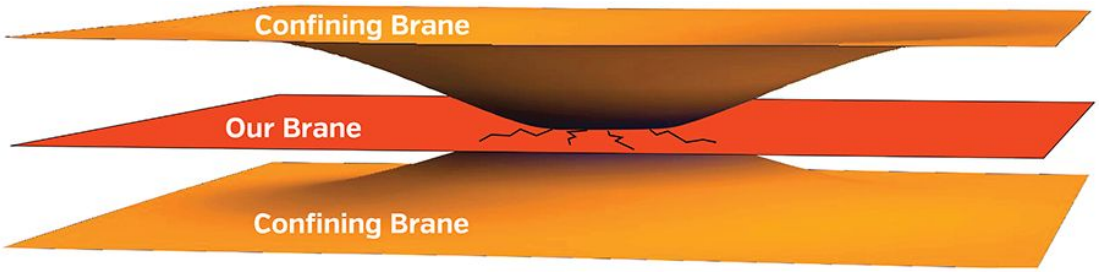


Fig. 25.6. Brane collision.

The Professor's Equation, at Last!

The laws of physics are expressed in the language of mathematics. Before Cooper met Professor Brand (in my extrapolation of the movie's story), the Professor tried to build a mathematical description of the bulk fields and how they might generate anomalies, control our universe's gravitational constant *G*, hold the wormhole open, and protect our brane from collisions.

In creating this mathematics, the Professor was guided by the trove of observational data his team was collecting (Chapter 24), and by Einstein's relativistic laws of physics in five dimensions.

The Professor embodied all his insights in a single equation, THE equation, which he wrote on one of the sixteen blackboards in his office (Figure 25.7).[44] Cooper sees the equation on his first visit to NASA, and the equation is still there thirty years later, when Murph has grown up to become a brilliant physicist in her own right, and is helping the Professor try to solve it.